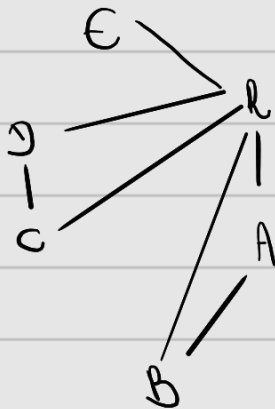


①

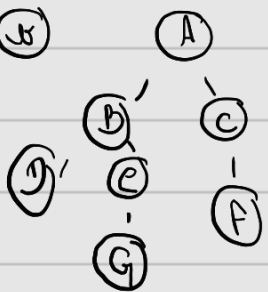


Não é possível, pois para fazer busca inteira
você se gasta precisando de 17 passos

②

a A-B-C-D-E-F-G

b



c A-B-E-G

③

a A-B-D-E-F-G-C



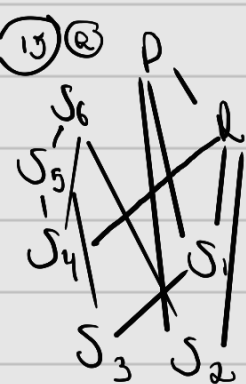
④

a 6

b B-D-G-E

③	A	B	C	D	E	F	G	H
inicial	∞	0	∞	∞	∞	∞	∞	∞
1º B	(1, B)	0	∞	(2, B)	∞	∞	(4, B)	(6, B)
2º A	(1, B)	0	(9, A)	(2, B)	∞	∞	(4, B)	(5, A)
3º D	(1, B)	0	(9, A)	(2, B)	∞	(4, D)	(3, D)	(5, A)
4º G	(1, B)	0	(9, A)	(2, B)	(6, G)	(4, D)	(3, D)	(5, A)
5º F	(1, B)	0	(5, F)	(2, B)	(6, G)	(4, D)	(3, D)	(5, A)
Final	(1, B)	0	(5, F)	(2, B)	(6, G)	(4, D)	(3, D)	(5, A)

⑭ não suintado = $\frac{n(n-1)}{2}$ suintado: $n(n-1)$



⑮ graus

P: 3	S3: 2
Q: 4	S4: 3
S1: 3	S5: 3
Sa: 3	S6: 3

⑮ não existe círculo ultramar,
pois para ser possível a
existência todos os graus
dos vértices devem ser pares.

⑯

⑮ a) Gráfico 1: não direcionado
Gráfico 2: direcionado

⑮ b) Gráfico 1: $\hat{n}$ ponderado, Gráfico 2: $\hat{n}$ ponderado

⑮ c) G1: simplex, G2: simplex

⑮ d) G1: conexo, G2: conexo

⑮ e) G1: não completo, G2: não completo

⑮ f) G1: não regular, G2: não regular

⑮ g) G1: sim: A-B-D-C-A, G2: sim: A-B-C-D-A

17) mesma da lista da sala		A	B	C	D	e	F	G
a) 0, 20, 40, 60, 80, 100, 120, 140	3 ^{ma}	∞	∞	∞	0	∞	∞	∞
b) 0, 5, 9, 20, 0, 3, 1, 2, 4	1 ^o	(5, 0)	(9, 0)	∞	0	(5, 0)	(1, 0)	∞
c) 0, 5, 4, 8, 0, 3, 1, 2, 4	2 ^o	(5, 0)	(9, 0)	∞	0	(4, 0)	(1, 0)	(2, 0)
d) D, F	3 ^o	(5, 0)	(9, 0)	∞	0	(3, 0)	(1, 0)	(2, 0)
	4 ^o	(5, 0)	(4, 0)	(8, 0)	0	(3, 0)	(1, 0)	(2, 0)

18)

a) não direcionadas, matriz de adjacência

b)



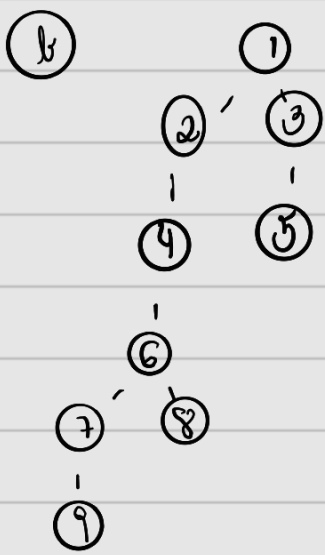
c) Componentes conexos: 2
ciclos: 1

d) A: B, C
B: A, C
C: A, B
D: e
e: D, F
F: e

19) 9-2-1-4-7-12

20)

a) 1-2-3-4-5-6-7-8-9

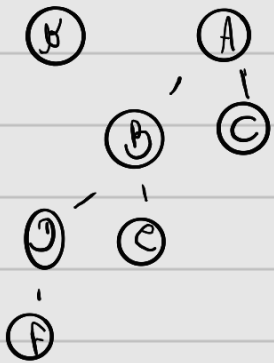


(c) 1-2-4-6-7-9

(d) 6 séries

(2)

(a) A-B-D-E-F-C



(c) $f \rightarrow e, e \rightarrow c$

(d) Não possui ciclo direcionado

(2)

(a) 0, 0, 0, 0, 0, 0, 0

(b) A-B-C-D-E-F

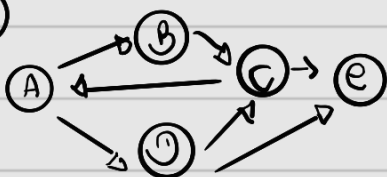
(c) 6

(d) A-B-C-E-F

	A	B	C	D	E	F
Ini	0	∞	∞	∞	∞	∞
1ª A	0	(2, A)	(5, C)	∞	∞	∞
2ª B	0	(2, A)	(3, B)	(5, D)	∞	(9, F)
3ª C	0	(2, A)	(3, B)	(5, D)	(5, E)	(9, F)
4ª D	0	(2, A)	(3, B)	(5, D)	(5, E)	(9, F)
5ª E	0	(2, A)	(3, B)	(5, D)	(5, E)	(6, F)
6ª F	0	(2, A)	(3, B)	(5, D)	(5, E)	(6, F)
FINAL	0	(2, A)	(3, B)	(5, D)	(5, E)	(6, F)

23 a) direcionado

b)



d) grau de entrada
 A: 1 C: 2 E: 2
 B: 1 D: 1

e) sim, existe, só percorrer pelas vertices: A → D → E

c) grau de saída

A: 2 C: 2 E: 0

B: 1 D: 2

24

a) não

b) 5: {A, B, C, D, E, F, G, H}

c) A, G, H

d) sim, D → E → F → G

e) grau

A: 0 C: 1 E: 2 G: 0

B: 1 D: 2 F: 2 H: 0

25

a) M → S

b) não existe, existe memória

c) graus

N: 5 R: 2

in: 1 un: 2

out: 4 out: 0

d) m: S, S

R: -

n: O, S, Q, R

S: m, m

O: m, m, n

P: R, P, m

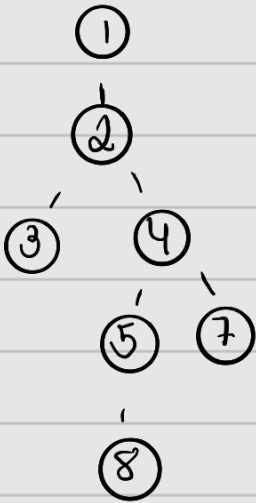
Q: -

e	m	n	O	P	Q	R	S
m	0	0	0	0	0	0	2
n	0	0	1	0	1	1	1
O	2	1	0	0	0	0	0
P	1	0	0	1	0	1	0
Q	0	0	0	0	0	0	0
R	0	0	0	0	0	0	0
S	2	0	0	0	0	0	0

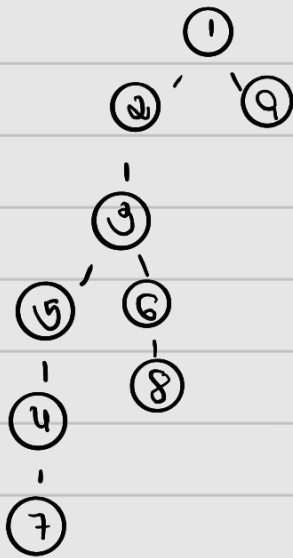
26

a) Graph 1: 1-2-3-4-5-7-8
Graph 2: 1-2-4-3-5-6-4-8-7

b) Graph 1:

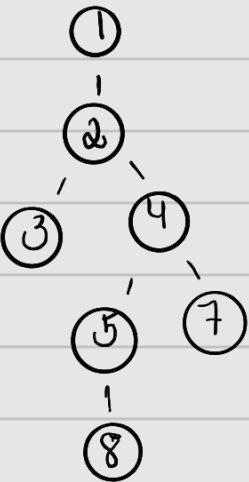


Graph 2:

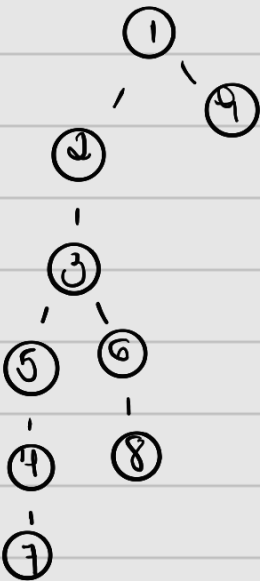


c) Graph 1: 1-2-4-7-5-8-3
Graph 2: 1-9-2-3-6-8-5-4-7

d) Graph 1:

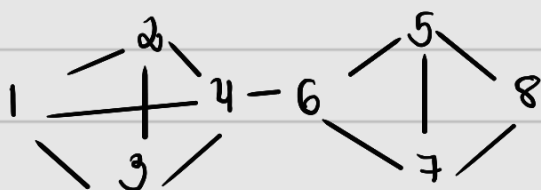


Graph 2:



27

a



b

	1	2	3	4	5	6	7	8
1	0	1	1	1	0	0	0	0
2	1	0	1	1	0	0	0	0
3	1	1	0	1	0	0	0	0
4	1	1	1	0	0	1	0	0
5	0	0	0	0	0	1	1	1
6	0	0	0	1	1	0	1	0
7	0	0	0	0	1	1	0	1
8	0	0	0	0	1	0	1	0

c

1: 2, 3, 4

2: 1, 3, 4

3: 1, 2, 4

4: 1, 2, 3, 6

5: 6, 7, 8

6: 4, 5, 7

7: 5, 6, 8

8: 5, 7

28

BFS:

A-B-F-C-D-H-G-E

DFS:

A-B-F-C-D-H-G-E

distância de A (BFS):

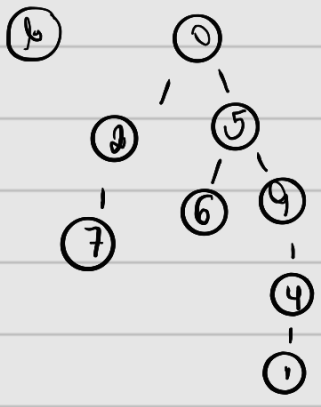
d(A): 0 d(D): 3 d(G): 2

d(B): 1 d(e): 1 d(H): 4

d(C): 2 d(F): 1

29

a) 0-2-5-7-6-9-4-1



c) 0-2-7-5-6-9-4-1



30

unício A

	A	B	C	D	E	F	G
INI	0	∞	∞	∞	∞	∞	∞
1ª	0	(5,A)	(3,A)	∞	∞	∞	∞
2ª	0	(5,A)	(3,A)	(10,C)	(10,C)	∞	∞
3ª	0	(5,A)	(3,A)	(10,C)	(8,B)	∞	(6,B)
4ª	0	(5,A)	(3,A)	(10,C)	(7,G)	∞	(6,B)
5ª	0	(5,A)	(3,A)	(9,E)	(7,G)	(8,E)	(6,B)
6ª	0	(5,A)	(3,A)	(9,E)	(7,G)	(8,E)	(6,B)
7ª	0	(5,A)	(3,A)	(9,E)	(7,G)	(8,E)	(6,B)
FINAL	0	(5,A)	(3,A)	(9,E)	(7,G)	(8,E)	(6,B)

a) d 0, 5, 3, 9, 7, 8, 6, 4

b) d -, A, A, e, G, e, B 3

c) A: A

B: A → B

C: A → C

G: A → B → G

E: A → B → G → E

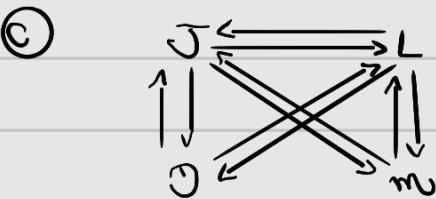
D: A → B → G → E → D

F: A → B → G → E → F

31

a) Jensen, Lenzinger, Dufur, modica

b) $J \rightarrow L, J \rightarrow D, J \rightarrow m$
 $L \rightarrow J, L \rightarrow D, L \rightarrow m$
 $D \rightarrow J, D \rightarrow L$
 $m \rightarrow J, m \rightarrow L$



d) $J: L, D, m$
 $L: J, D, m$
 $D: J, L$
 $m: J, L$

	J	L	D	m
J	0	1	1	1
L	1	0	1	1
D	1	1	0	0
m	1	1	0	0

32

	0	1	2	3	4	5
INICIO	∞	∞	∞		∞	0
1º J	(2,5)	(4,5)	∞	(6,5)	∞	0
2º D	(2,5)	(4,5)	∞	(6,5)	(17,0)	0
3º L	(2,5)	(4,5)	∞	(6,5)	(17,0)	0
4º 3	(2,5)	(4,5)	(18,3)	(6,5)	(17,0)	0
5º 4	(2,5)	(4,5)	(18,3)	(6,5)	(17,0)	0

6°2	(2,5)	(4,5)	(18,3)	(6,5)	(17,0)	0
FINAL	(2,5)	(4,5)	(18,3)	(6,5)	(17,0)	0

(b) 5-3-2

(c) 18

(33)

(a)	1	2	3	4	5	
1	0	0	0	1	2	1: 4, 5, 5
2	0	1	1	1	1	2: 2, 3, 4, 5
3	0	1	0	1	0	3: 2, 4
4	1	1	1	0	0	4: 1, 2, 3
5	2	1	0	0	0	5: 1, 1, 2

(b)	1	2	3	4	5	6	7	
1	0	1	1	0	0	0	0	1: 2, 3
2	1	0	1	1	0	1	1	2: 1, 3, 4, 6, 7
3	1	1	0	0	1	1	1	3: 1, 2, 5, 6, 7
4	0	1	0	0	0	1	0	4: 2, 6
5	0	0	1	0	0	0	1	5: 0, 7
6	0	1	1	1	0	0	1	6: 2, 3, 4, 7
7	0	1	1	0	1	1	0	7: 2, 3, 5, 6

(c)	1	2	3	4	
1	2	3	1	0	1: 2, 2, 2, 3
2	0	1	1	1	2: 2, 3, 4
3	0	0	0	1	3: 4
4	0	0	0	0	4: -

